Number of Subset Xors

Time limit: 1.00 s **Memory limit:** 512 MB

Given an array of n integers, your task is to find the number of different subset xors.

Input

The first line has an integer n: the size of the array.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the number of different subset xors.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $0 \le x_i \le 10^9$

Example

Input	Output
3 3 6 5	4

Explanation: The following values can be the xor of a subset:

- $0 = \text{xor of the empty set}$
- $3 = 3$
- $5 = 3 \oplus 6$
- $6 = 3 \oplus 5$

In this case, no other values can be the xor of a subset.